PassFrame tuition

Statistics & Averages

Year 4 Pt 2 — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: Find the mean of 11, 15, 13, 17, 9.
Method & answer: 11+15+13+17+9=65, 65÷5=13. Answer: 13.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the mean of 8, 10, 12, 14, 6.

2. Find the mean of 13, 15, 17.

3. Find the median of 9, 15, 7, 13, 11.

4. Find the median of 20, 8, 26, 14, 22.

5. Find the mode of 6, 10, 10, 13, 5, 10.

6. Find the mode of 12, 12, 16, 16, 16, 7.

7. Find the range of 11, 19, 14, 25, 16.

8. Find the range of 60, 45, 75, 40.

9. Which of these is the mode of 8, 8, 11, 14, 14, 14? A) 14 B) 8 C) 11 D) 12

10. A table shows pets: Dogs=13, Cats=9, Fish=7, Birds=5. How many pets are there in total?

11. Cakes sold over 4 days: Mon=26, Tue=35, Wed=21, Thu=30. What is the mean number sold per day?

12. Select ALL that apply. A frequency table shows: 1 pet = 9 pupils, 2 pets = 12 pupils, 3 pets = 7 pupils. Which statements are true? A) 28 pupils were surveyed in total B) The mode number of pets owned is 2 C) The mode number of pets owned is 1 D) More pupils own 2 pets than own 3 pets

13. Five friends scored 24, 30, 33, 27, 26 in a test. What is their mean score? A) 28 B) 27 C) 29 D) 30

14. Find the mean of 7, 7, 7, 7, 12.

15. Find the range of 16, 31, 23, 8, 19.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the median of 11, 16, 8, 13.

2. Find the mean of 21, 25, 17, 29.

3. A pictogram shows books read: Amy=11, Ben=15, Cara=8, Dan=18. How many books were read

6. A bar chart shows sales at 4 shops: A=23, B=31, C=16, D=26. What is the mean sale per shop?

7. Find the mode of 5, 8, 8, 10, 12, 8.

8. A table shows pets: Dogs=19, Cats=15, Fish=11, Birds=8. If 7 more fish are added, how many pets are

11. Which of these is the range of 43, 21, 35, 54, 29? A) 33 B) 54 C) 21 D) 32

12. Find the median of 11, 6, 15, 9, 19, 12.

13. A pie chart shows a survey of 44 pupils: half prefer football. How many pupils prefer football?

altogether?

there altogether now?

4. Which of these is the median of 10, 19, 14, 27, 7? A) 14 B) 19 C) 10 D) 27

9. Select ALL that apply. A class recorded favourite subjects: Maths=17, Art=12, Science=9. Which statements are true? A) 38 pupils were asked in total B) Maths is the mode (most popular) C) Science is the mode (most popular) D) More pupils chose Art than Maths

14. Which of these is the mean of 20, 24, 28, 32? A) 26 B) 28 C) 24 D) 30

5. Find the range of 26, 54, 39, 17, 46.

10. Find the mean of 13, 19, 15, 21.

15. Find the range of 75, 100, 87, 83.

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A café's mean daily takings over 5 days is £65. What was the total takings over the 5 days?

6. A team's goals in 6 matches were 3, 1, 4, 2, 5, 3. Find the mean, median and mode.

11. A trail-mix recipe (360° pie chart, total 720g) shows: nuts = 140°, raisins = 100°, seeds = 60°, and the rest is chocolate. What is the weight of chocolate, in grams?

2. The mode of a set of numbers is 12, and the range is 7. If the smallest number is 12, what is the largest number?

7. A survey of 60 pupils' travel is shown in a pie chart: half walk, a fifth cycle, and the rest come by car. How many travel by car?

12. The mean of 5 numbers is 28. Four of the numbers are 22, 26, 30 and 34. What is the fifth number?

3. The mean of four numbers is 19. Three of the numbers are 14, 20 and 22. What is the fourth number?

8. A set of 7 numbers has a mean of 22. If one of the numbers, 10, is removed, what is the mean of the remaining 6 numbers? A) 24 B) 22 C) 25 D) 23

13. A bag of 8 numbers has a mean of 12. Which of these MUST be true? A) The total of the 8 numbers is 96 B) Every number in the bag is 12 C) The largest number is greater than 12 D) The smallest number is less than 6

4. A pie chart is split into just two colours so that green is four times as big as yellow. What is the angle for yellow?

9. A pie chart (360°) shows how 120 pupils travel to school: Walk = 180°, Car = 90°, Bus = 90°. How many pupils travel by car or bus combined?

14. Two classes of 20 pupils each recorded a mean of 1.5 pets per pupil in Class A and 0.9 pets per pupil in Class B. What is the mean number of pets per pupil across both classes combined?

5. Five numbers, in order, are 15, 19, ?, 29, 35, and their median is 24. What is the missing middle number?

10. Select ALL that apply. A class of 25 pupils' test scores have mean 58 and range 36. Which statements MUST be true? A) The total of all 25 scores is 1,450 B) At least one score is below 58 C) The highest score minus the lowest score is 36 D) Every pupil scored above 50

15. A pie chart shows a class's eye colour: Brown = 180°, Blue = 120°, Green = 60°. If there are 42 pupils, how many more have brown eyes than blue eyes?